

Spiking Transformer with Spatial-Temporal Attention

CVPR 2025

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Outline

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- 2 Proposed Method: STAtten
- 3 Experimental Results
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Spiking Neural Networks (SNNs) – Background

- **Bio-inspired** computing paradigm
- Communicate via **sparse binary spikes** instead of continuous activations
- Significant **energy savings** on neuromorphic hardware (TrueNorth, Loihi, Tianjic)
- Fundamental neuron model: **Leaky Integrate-and-Fire (LIF)**

LIF Neuron Dynamics

$$\mathbf{u}[t+1]' = \tau \mathbf{u}[t]' + \mathbf{W}' f(\mathbf{u}[t]'^{-1})$$

$$f(\mathbf{u}[t]') = \begin{cases} 1 & \text{if } \mathbf{u}[t]' > V_{\text{th}} \\ 0 & \text{otherwise} \end{cases}$$

τ : leaky factor V_{th} : threshold

The Surrogate Gradient Problem and Solution

The Challenge: Spike function is non-differentiable

$$f(\mathbf{u}[t]') = \begin{cases} 1 & \text{if } \mathbf{u}[t]' > V_{\text{th}} \\ 0 & \text{otherwise} \end{cases}$$

The derivative is:

$$\frac{df}{d\mathbf{u}} = \begin{cases} \text{undefined} & \text{at } \mathbf{u} = V_{\text{th}} \\ 0 & \text{elsewhere} \end{cases}$$

Problem: Zero gradient everywhere $\Rightarrow$ **no learning with backpropagation!**

Solution: Surrogate Gradient Method

Replace the true (zero) gradient with a smooth approximation during backpropagation:

$$\frac{df}{d\mathbf{u}} \approx \sigma'(\mathbf{u} - V_{\text{th}}) = \frac{1}{(1 + |\mathbf{u} - V_{\text{th}}|)^2}$$

Forward Pass

Compute true binary spikes: $f(\mathbf{u}) \in \{0, 1\}$

Backward Pass

Use smooth surrogate gradient for weight updates

Key insight: The surrogate is **never evaluated** on actual spikes — it only affects gradient computation. This enables SNNs to be trained end-to-end with backprop while maintaining spike sparsity.

Mathematical Derivation: Why Gradient = 0

Step 1: Recall the spike function (Heaviside step)

$$f(u) = H(u - V_{th}) = \begin{cases} 1 & \text{if } u > V_{th} \\ 0 & \text{if } u \leq V_{th} \end{cases}$$

Step 2: Compute the derivative using definition

By definition, the derivative is:

$$\frac{df}{du} = \lim_{\Delta u \rightarrow 0} \frac{f(u + \Delta u) - f(u)}{\Delta u}$$

Step 3: Analyze three regions

Region 1: $u \ll V_{th}$

$$\begin{aligned} f(u) &= 0 \\ f(u + \Delta u) &= 0 \\ \frac{\Delta f}{\Delta u} &= \frac{0 - 0}{\Delta u} = 0 \end{aligned}$$

$$\Rightarrow \boxed{\frac{df}{du} = 0}$$

Region 2: $u = V_{th}$ (threshold)

$$\begin{aligned} \text{LHS: } \lim_{\Delta u \rightarrow 0^-} \frac{0 - 0}{\Delta u} &= 0 \\ \text{RHS: } \lim_{\Delta u \rightarrow 0^+} \frac{1 - 0}{\Delta u} &= \infty \\ \Rightarrow \boxed{\text{Undefined!}} \end{aligned}$$

Region 3: $u \gg V_{th}$

$$\begin{aligned} f(u) &= 1 \\ f(u + \Delta u) &= 1 \\ \frac{\Delta f}{\Delta u} &= \frac{1 - 1}{\Delta u} = 0 \end{aligned}$$

$$\Rightarrow \boxed{\frac{df}{du} = 0}$$

Step 4: Conclusion

$$\frac{df}{du} = \begin{cases} 0 & \text{if } u < V_{th} \text{ (not spiking)} \\ \text{undefined} & \text{if } u = V_{th} \text{ (threshold)} \\ 0 & \text{if } u > V_{th} \text{ (spiking)} \end{cases}$$

× **Result: Gradient is 0 everywhere (except undefined at V_{th})**

This is why backpropagation fails: $\frac{\partial \mathcal{L}}{\partial W} = \frac{\partial \mathcal{L}}{\partial s} \cdot 0 \cdot \dots = 0$

Why Transformers for SNNs?

- Convolution-based SNNs suffer from **information loss** due to binary activations
- Transformers capture **long-range dependencies** via self-attention
- Spike-based transformers replace Softmax with **binary Q, K, V** operations

Key architectures:

- Spikformer (2023)
- Spike-driven Transformer (2023)
- SpikingReformer, QKFormer (2024)

Spike-based Self-Attention (SSA)

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} = \text{LIF}(\mathbf{W}\mathbf{X})$$

$$\text{SA} = \text{LIF}(\mathbf{Q}\mathbf{K}^{\top}\mathbf{V} \cdot \alpha)$$

No Softmax $\Rightarrow$ efficient binary computation

Critical Limitation

All existing methods use **spatial-only attention**, ignoring temporal dynamics.

The Problem: Spatial-Only Attention Misses Temporal Information

Spatial-Only Attention

- Operates **independently** at each timestep t
- Captures vertical (spatial) relationships only
- Misses features that **evolve over time**

$$S_Atn[t] = LIF(Q[t]K^T[t]V[t] \cdot \alpha)$$

Spatial-Temporal Attention (Proposed)

- Jointly captures **spatial + temporal** correlations
- Attention spans across timesteps
- Better feature representation

$$ST_Atn[:, d] = LIF(Q[:, d]K^TV \cdot \alpha)$$

Key insight from entropy analysis: Spatial-temporal attention achieves **lower entropy** (4.85 vs 5.81) and **higher accuracy** (79.9% vs 77.7%) on CIFAR-100 compared to spatial-only attention.

Self-Supervised Learning and SNNs

Current challenge: Most SNNs are trained with **supervised learning** (surrogate gradients), requiring large labeled datasets.

Why SSL matters for SNNs:

Pretraining without labels

- Contrastive or masked objectives adapted to spike-based representations
- Pretrain on large unlabeled datasets

Transfer to downstream

- Pretrain on large unlabeled data
- Fine-tune for detection/segmentation with few labels

Better representations

SSL forces encoders to learn robust, noise-invariant features — which aligns perfectly with inherent noise in spike-based computing.

Key challenge

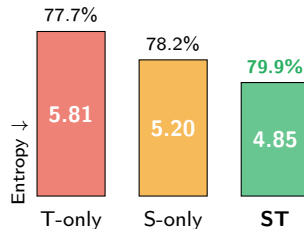
Adapting SSL to binary spike activations — cannot use pixel-level reconstruction. Joint-embedding methods (latent-space objectives) are more compatible with SNNs.

Takeaway: STAtten's improved temporal modeling enables **richer representations**, creating a foundation for future SSL+SNN approaches.

Shannon Entropy Analysis

$$H(\text{Attn}) = - \sum_{t,n,d} \hat{p}[t, n, d] \log \hat{p}[t, n, d]$$

- Lower entropy $\Rightarrow$ more focused spike patterns
- Higher entropy $\Rightarrow$ dispersed, less informative



Results (SDT on CIFAR-100):

Inverse correlation: lower entropy $\longleftrightarrow$ higher accuracy $\Rightarrow$ spatial-temporal attention produces more structured representations

Challenges of Full Spatial-Temporal Attention

Why not simply correlate all timesteps?

Challenge 1: Memory

- Full temporal attention requires storing all timestep information simultaneously
- Matrix multiplication over $(T \times N, D)$
- Leads to $O(T^2 N^2 D)$ complexity

Challenge 2: Dead Neurons

- Binary Q, K, V $\Rightarrow$ distant timesteps have weak spike similarity
- **Higher active neurons** for same-timestep pairs
- Activity **decreases** with temporal distance

Solution: Block-wise partitioning of temporal sequences
 $\Rightarrow$ captures *local* temporal correlations efficiently

STAtten: Block-Wise Spatial-Temporal Attention

Core Formulation

Partition temporal sequence into blocks of size B :

$$\text{STAtten}(\mathbf{X}[b]) = \text{LIF}\left(\mathbf{Q}[b] \mathbf{K}^\top[b] \mathbf{V}[b] \cdot \alpha\right)$$

where $[b] = [iB : (i+1)B, :, d]$, $i \in \{0, \dots, T/B - 1\}$

Efficient Computation via Reshaping:

$$\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{\frac{T}{B} \times N \times D}$$
$$\xrightarrow{\text{reshape}} \mathbb{R}^{(\frac{T}{B} N) \times D}$$

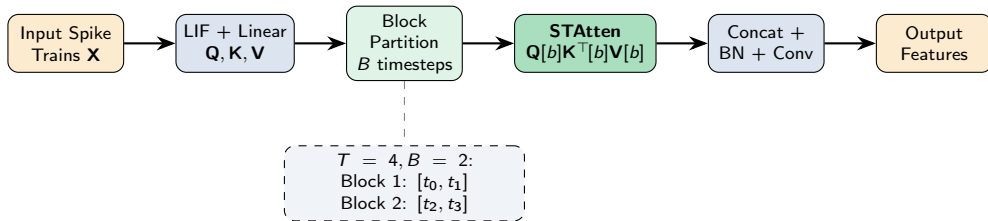
$$\mathbf{K}^\top \mathbf{V} \in \mathbb{R}^{D \times D}$$

$$\mathbf{Q}(\mathbf{K}^\top \mathbf{V}) \in \mathbb{R}^{\frac{T}{B} \times N \times D}$$

Key Properties

- Complexity: $O(TND^2)$ – **same** as spatial-only
- No Softmax $\Rightarrow$ flexible ordering
- $\sim 1.6\times$ higher batch size vs full temporal
- $B = 2$ (static), $B = 4$ (neuromorphic)

Architecture Overview



Plug-and-play: STatten replaces only the self-attention module. Residual connections remain unchanged. Compatible with Spikformer, SDT, SDT-V2, SpikingReformer, QKFormer.

Integration with Existing Architectures

SEW Shortcut (Spikformer, QKFormer)

- Spike activations added directly to attention output
- Element-wise residual addition

MS Shortcut (SDT, SDT-V2, SpikingReformer)

- Propagates **membrane potentials** instead of spikes
- Richer gradient flow

Table: *

Attention Mechanisms Replaced	
Architecture	Original → STAtten
Spikformer	SSA → STAtten
SDT	SDSA → STAtten
SDT-V2	SDSA → STAtten
SpikingReformer	DSSA → STAtten
QKFormer	SSA → STAtten

$$\mathbf{F}_{\text{out}} = \text{Concat}\left(\text{STAtten}(\mathbf{X}[iB : (i+1)B])_{i=0}^{T/B-1}\right) \rightarrow \mathbf{Z} = \text{BN}(\text{Conv}(\mathbf{F}_{\text{out}}))$$

Computational Complexity and Energy

Method	Type	Complexity	Energy
ViT	ANN	$O(N^2 D)$	$E_{\text{MAC}} \cdot N^2 D$
ViViT	ANN	$O(T^2 N^2 D)$	$E_{\text{MAC}} \cdot T^2 N^2 D$
Spikformer	SNN	$O(TND^2)$	$E_{\text{AC}} \cdot TND^2(S_Q + S_K + S_V)$
SDT	SNN	$O(TND)$	$E_{\text{AC}} \cdot TND(S_Q + S_K)$
QKFormer	SNN	$O(TND^2)$	$E_{\text{AC}} \cdot TND^2(S_Q + S_K + S_V)$
STAtten	SNN	$O(TND^2)$	$E_{\text{AC}} \cdot TND^2(S_Q + S_K + S_V)$

Key Takeaway

STAtten has **identical complexity and energy** to spatial-only spiking transformers, but captures richer features.

Why?

- No Softmax $\Rightarrow$ flexible $\mathbf{K}^T \mathbf{V}$ first
- AC ops vs MAC ops
- Energy modulated by firing rates S_Q, S_K, S_V

Why No Softmax? Complexity Derivation

Standard Transformer Attention: $\text{Attention} = \text{Softmax}(QK^\top / \sqrt{d}) V$

Softmax forces $QK^\top \in \mathbb{R}^{N \times N}$ first $\Rightarrow \mathbf{O}(N^2 D)$ per timestep.

Key insight: Binary spikes $\Rightarrow$ no Softmax $\Rightarrow$ associative reordering:

$$\text{STAtten} = \mathbf{Q} \cdot (\mathbf{K}^\top \mathbf{V})$$

Spatial-Only (per timestep)

$$\mathbf{K}^\top \mathbf{V} : O(ND^2)$$

$$\mathbf{Q} \cdot (\mathbf{K}^\top \mathbf{V}) : O(ND^2)$$

Over T timesteps: $\mathbf{O}(TND^2)$

STAtten (per block of B timesteps)

Tokens: BN

$$\mathbf{K}^\top \mathbf{V} : O(BND^2)$$

$$\mathbf{Q} \cdot (\mathbf{K}^\top \mathbf{V}) : O(BND^2)$$

Over T/B blocks: $\mathbf{O}(TND^2) \checkmark$

The B factors **cancel out**. Complexity is identical.

Sequential CIFAR-10/100 – Temporal Dependency Test

Setup:

- Images divided **column-wise**: each column = one timestep
- Sequence length: 32 timesteps
- Tests **long-range temporal dependencies**

Method	s-C10	s-C100
PLIF	81.47	53.38
Spikformer	79.29	57.17
+ STAtten	82.45	58.75
SDT	80.32	61.08
+ STAtten	83.41	64.30
SpikingReformer	79.25	57.48
+ STAtten	81.84	58.30

Up to **+3.09%** (s-CIFAR10) and **+3.22%** (s-CIFAR100) improvement

CIFAR-10/100 – Static Image Classification

Method	Architecture	T	C10 (%)	C100 (%)
tdBN	ResNet19	4	92.92	–
PLIF	ConvNet	8	93.50	74.24
Dspike	ResNet18	6	94.25	78.50
DSR	ResNet18	20	95.40	–
Spikformer	Spikformer-4-384	4	93.99	75.06
+ STAtten		4	94.36	75.85
SDT	SDT-2-512	4	95.60	78.40
+ STAtten		4	96.03	79.85
QKFormer	QKFormer-4-384	4	95.12	79.79
+ STAtten		4	95.35	80.20

Consistent improvement across **all** backbone architectures – trained from scratch.

ImageNet – Large-Scale Classification

Method	Type	Params	T	Acc (%)
ViT-B/16	ANN	86.59M	1	77.90
DeiT-B	ANN	86.59M	1	81.80
Spikformer-8-768	SNN	66.34M	4	74.81
QKFormer-6-1024	SNN	64.96M	4	84.22
SDT-8-768	SNN	66.34M	4	76.32 / 77.07*
+ STAtten	SNN	66.34M	4	78.11 / 78.39*
SDT-V2-8-512	SNN	55.4M	4	79.49 / 79.98*
+ STAtten	SNN	55.4M	4	79.85 / 80.67*

*288×288 resolution. STAtten + SDT-V2 surpasses larger models with **20% fewer parameters**.

Neuromorphic Datasets – CIFAR10-DVS & N-Caltech101

Datasets:

- **CIFAR10-DVS**: 10k samples, Dynamic Vision Sensor, 16 timesteps
- **N-Caltech101**: 8,831 DVS samples, 101 classes
- Resolution: 64×64
- Block size $B = 4$

Method	DVS (%)	N-Cal (%)
NDA	79.6	–
SDT	80.0	81.80
+ STAtten	81.1	83.15
SpikingReformer	78.80	81.29
+ STAtten	80.60	81.95
QKFormer	82.90	83.58
+ STAtten	83.90	84.25

State-of-the-art: 83.90% on CIFAR10-DVS & 84.25% on N-Caltech101

Memory and Energy Efficiency on ImageNet

Method	Precision	Mem (MB)	Energy (mJ)	Acc (%)
ViT-B/16	32/32	351.8	254.84	77.90
Liu (ViT-B)	8/8	87.96	110.8	–
Spikformer	32(8)/1	285.99 (86.48)	21.48	74.81
QKFormer	32(8)/1	280.28 (84.99)	38.91	84.22
SDT	32(8)/1	283.97 (87.42)	12.42	76.32
STAtten+SDT	32(8)/1	283.97 (87.42)	12.36	78.11
SDT-V2	32(8)/1	250.22 (84.02)	52.40	79.49
STAtten+SDT-V2	32(8)/1	250.22 (84.02)	52.38	79.85

STAtten: **higher accuracy, same energy & memory.**
SNN: ~ 12 mJ vs ANN: ~ 255 mJ ($20\times$ reduction)

CIFAR-100

- Consistent $\sim 0.5\text{--}1.0\%$ improvement across model sizes (2.56M to 22.97M params)
- Improvement maintained at **all scales**

Sequential CIFAR-100

- Larger improvement: $\sim 3\text{--}5\%$
- Temporal dependencies are more critical in sequential tasks
- Robust gains even with **smaller architectures**

Key finding: STAtten's benefit **does not require larger models**.
The spatial-temporal attention enriches representations at **every scale**,
but the gain is most pronounced for **temporally-rich tasks**.

Experimental Results: Our Work vs Paper

Method	Config	T	CIFAR-10	CIFAR-100
SDT (Baseline)	SDT-2-256	2	87.12%	61.92%
STAtten (Paper)	SDT-2-512	4	96.03%	79.85%
STAtten (Ours)	SDT-2-256	2	88.72%	66.70%
MSB-STAtten (Ours)	SDT-2-256	2	88.91%	[in progress]

Our Improvements

- STAtten: **+1.60%** CIFAR-10
- STAtten: **+4.78%** CIFAR-100
- MSB-STAtten: **+0.19%** over STAtten
- **Zero** extra parameters

Why the gap vs Paper?

- Paper: larger backbone (512 vs 256)
- Paper: T=4 vs T=2 (GPU memory)
- **But:** our relative gains validate the approach
- Temporal modeling works even with limited resources

Key insight: STAtten's gains are **3× larger on CIFAR-100** (+4.78%) than CIFAR-10 (+1.60%), showing temporal modeling is especially valuable on complex tasks.

Current Limitations

- **Hardware deployment** on neuromorphic chips is challenging
- These chips process info **step-by-step**, but STAtten needs block-level temporal access
- Full spatial-temporal attention is impractical for deployment

Potential Solutions

- **Layer-by-layer chips** that support layer-wise processing across timesteps
- **Parallel LIF neurons** for accelerated computation
- Block-wise approach is a step toward practical deployment

Open Questions

- Optimal block size B selection
- Extension to NLP, audio
- Larger neuromorphic datasets

Summary of Contributions

1. **Problem identification:** Entropy analysis shows spatial-only attention misses crucial temporal dependencies
2. **STAtten mechanism:** Block-wise spatial-temporal attention maintaining $O(TND^2)$ complexity – **no additional overhead**
3. **Plug-and-play design:** Seamless integration into 5 existing spiking transformer architectures
4. **Consistent improvements:** SOTA on both static (CIFAR, ImageNet) and neuromorphic (DVS, N-Caltech) datasets with **similar energy**

Core message: Temporal information matters in spiking transformers.
STAtten captures it efficiently with **zero overhead**.

Key Results at a Glance

Benchmark	Best STAtten	Improvement
Sequential CIFAR-10	83.41%	+3.09%
Sequential CIFAR-100	64.30%	+3.22%
CIFAR-10	96.03%	+0.43%
CIFAR-100	80.20%	+1.45%
ImageNet (224)	79.85%	+0.36%
ImageNet (288)	80.67%	+0.69%
CIFAR10-DVS	83.90%	+1.00%
N-Caltech101	84.25%	+0.67%

All improvements with **same complexity** $O(TND^2)$ and **similar energy**.

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Thank You

Questions?

Code: `github.com/Intelligent-Computing-Lab-Yale/STatten`

Paper: Lee et al., CVPR 2025